Zewail City - University of Science and Technology **SCHOOL OF COMPUTATIONAL SCIENCES AND ARTIFICIAL INTELLIGENCE** **CSAI 201: Data Structures and Algorithms**	Fall 2025

Data Structures & Algorithms

Smart City: Traffic Management

Project Requirements Document

Introduction

Modern smart cities require intelligent traffic management to balance efficiency, safety, and fairness. Traffic signals at intersections must allocate green light slots fairly to different vehicle types, while giving priority to emergency vehicles. Congestion, unexpected events, and fairness among road users are all critical challenges.

This project simulates the **Traffic Control Center (TCC)** of a smart city. Vehicles arrive dynamically at multiple intersections, each with multiple lanes and traffic lights. The TCC must decide which lane receives the green light at each timestep, considering priorities, waiting times, and fairness rules. Beyond intersections, this simulation also considers **multi-city zones**, **interconnected roads**, and **adaptive policies**.

Your simulation should:

1. Assign green light slots according to the defined priority rules.
2. Handle dynamic events such as arrivals, cancellations, promotions, accidents, and road closures.
3. Generate a comprehensive output file with detailed traffic statistics.
4. Support multiple simulation modes: interactive, step-by-step, and silent.
5. Scale up to handle **thousands of vehicles across tens of intersections**.

Definitions

- **AT (Arrival Time):** Time when a vehicle enters the system.
- **WT (Waiting Time):** Time spent waiting since arrival until served.
- **XD (Crossing Duration):** Time needed for a vehicle to cross once green is granted.
- **CT (Completion Time):** `CT = AT + WT + XD.`
- **AP (Auto Promotion Timesteps):** Maximum waiting time for PT (Public Transport) before automatic promotion to EV.
- **CancelT (Cancellation Threshold):** Maximum waiting time for NC (Normal Cars) before canceling.
- **Signal Switch:** A change from one green lane to another (incurs switching cost).
- **Intersection Utilization:** % of time an intersection actively serves vehicles.
- **Accident Event:** An unexpected blockage that reduces available lanes temporarily.
- **Zone Utilization:** A higher-level metric for groups of intersections.

Entities

1. Vehicles

Each vehicle request includes the following fields, which must be loaded from the input file and maintained throughout the simulation:

- **Arrival Time (AT):** timestep when the vehicle arrives at the intersection.
- **ID (Vehicle ID):** A unique integer identifier assigned to every vehicle.
- **Vehicle Type:**
    - **Emergency Vehicles (EV):** Highest priority. Can interrupt green. (e.g., ambulances, fire trucks).
    - **Public Transport (PT):** Medium priority. Promoted to EV if WT ≥ AP. (e.g., buses, trams).
    - **Normal Cars (NC):** Lowest priority. May cancel if WT ≥ CancelT.
    - **Freight Vehicles (FV):** Large vehicles with longer crossing durations. Medium-low priority, but cannot be interrupted once crossing.

- **Intersection (INT):** The numeric identifier of the intersection at which the vehicle is queued.
- **Lane ID (LN):** The lane where the vehicle is queued.
- **Crossing Duration (XD):** The number of timesteps the vehicle requires to clear the intersection once it receives a green signal. Larger vehicles (e.g., FV) usually have higher XD.
- **Priority Score (PR) [Optional]:** For EV only, a numeric score representing urgency, criticality of service, or mission priority. Higher PR increases its scheduling chance when multiple EVs are present.
- **CargoFlag [Optional]:** For FV only, a binary indicator specifying whether the freight vehicle is carrying high-priority cargo. If set, the vehicle may be elevated to higher priority handling in specific zones (e.g., highways).

2. Intersections

- The city has **I intersections** (loaded from input file).
- Identified by IDs (1 … I).
- Each intersection has **multiple incoming lanes**.
- Only one lane may be active at a time.
- Intersections may be grouped into **zones** (downtown, suburban, highway) to test large-scale performance.
- Policies may differ by zone (e.g., suburban may relax AutoP rules).
- A **queue** of waiting vehicles represents each lane. First-Come-First-Served for PT and NCs. EVs may override.
- Only one lane can be active (green) at a time per intersection.

3. Traffic Lights

- Each intersection has traffic lights that switch between lanes.
- Switching incurs a **switching cost (delay in seconds)**. Must minimize switching delays while respecting priorities.
- The Traffic Control Center decides the **active lane** each timestep using assignment rules.

Simulation Time (Timesteps)

The simulation runs in discrete **timesteps**, where one timestep represents a single unit of simulation time (e.g., one second). During each timestep:

- All new vehicle arrival events scheduled for this timestep are executed.
- The Traffic Control Center (TCC) evaluates lane priorities and may switch the green signal.
- Vehicles currently crossing reduce their remaining crossing duration by 1 timestep.
- Completed vehicles exit the system, and their completion times are recorded.
- Accidents and road closures take effect or expire if their scheduled duration is reached.

The simulation advances timestep by timestep until there are no more active events and all vehicles have either crossed successfully, canceled, or been rerouted and finished.

Assignment Criteria

The assignment criteria define how the Traffic Control Center (TCC) chooses which lane and vehicle to serve at each timestep. This is the heart of the scheduling logic and ensures fairness while respecting emergency priorities.

- **EV (Emergency Vehicles):** Always served first. They can interrupt the current green signal. Among multiple EVs, the selection is determined by the following formula:

  Priority = (Urgency × 2) − WT

  Where *Urgency* is a field given in the input (or derived from PR), *WT* is the waiting time. The higher the score, the earlier the EV should be served.

- **PT (Public Transport):** Served only if no EV is waiting. PT vehicles follow FCFS (First-Come-First-Served) within each lane. However, if a PT vehicle's waiting time exceeds the AutoP threshold (AP), it is automatically promoted to EV status and treated with higher priority.

- **NC (Normal Cars):** These vehicles are the default lowest priority. They are served in FCFS order only when no EVs or PTs are waiting. If an NC exceeds the cancellation threshold (CancelT), it cancels and is removed from the system.
- **FV (Freight Vehicles):** FVs are handled specially because they require longer crossing durations and occupy more road space. Once an FV begins crossing, it locks the lane until completion. They are generally considered lower than PT but above NC, unless carrying high-priority cargo, in which case their priority can be temporarily raised.

Events

Events are the driving force of the simulation. Each event occurs at a specific timestep and alters the state of the system. Below are the supported event types explained in detail:

Arrival (A)

Format: `A TYPE AT ID INT LN XD [PR]`

This event introduces a new vehicle into the system. TYPE can be EV, PT, NC, or FV. The fields specify when the vehicle arrives (AT), its unique ID, intersection (INT) and lane (LN), crossing duration (XD), and optional priority score for EVs. When an arrival event occurs, the vehicle is added to the designated lane queue. If an EV arrives, it may preempt the current green signal immediately.

Cancellation (X)

Format: `X AT ID`

This event represents a vehicle canceling its request due to excessive waiting. Typically, it applies to NCs, but depending on thresholds, others may also be canceled. At time (AT), the system removes the vehicle with the given ID if it is still in a waiting queue. If the vehicle has already crossed, the cancellation is ignored. Cancellations directly affect statistics, as they increase the canceled count.

Promotion (P)

Format: `P AT ID`

This event upgrades a PT vehicle into an EV. Promotion may be triggered manually via the input file or automatically when a PT's waiting time exceeds the AutoP threshold. Once promoted, the PT is treated exactly like an EV and may immediately preempt the lane assignment. Promotions ensure that PT vehicles are not starved.

Accident (ACC)

Format: `ACC AT INT LN DUR`

This event simulates an accident that blocks a specific lane of an intersection. The lane becomes unavailable at time AT and remains blocked for DUR timesteps. Vehicles in that lane cannot move forward until the blockage clears. During the accident, the system may reroute vehicles if alternative paths are available; otherwise, they must wait.

Road Closure (RC)

Format: `RC AT INT DUR`

This event closes an entire intersection. All its lanes become unavailable for DUR timesteps starting at AT. Vehicles waiting in those lanes must be rerouted if rerouting is enabled. Road closures test the robustness of the simulation by simulating large-scale disruptions.

Event Handling Logic

- Arrival adds a vehicle to the proper queue.
- Cancellation removes a vehicle from the queue if still waiting.
- Promotion changes a PT into an EV, potentially altering scheduling.
- An accident marks a lane as blocked, preventing movement until the specified duration expires.
- Road closure marks the entire intersection as unavailable. Vehicles may be rerouted or blocked depending on the configuration.

Rerouting During ACC and RC

When a lane (ACC) or an entire intersection (RC) is blocked, the system may attempt **rerouting** if this feature is enabled. Rerouting works as follows:

- Each vehicle has a target intersection. If its assigned lane or intersection is closed, the system checks for alternative intersections that can serve the vehicle type.
- A **rerouting map** or adjacency list of intersections (representing city road topology) must be defined at the beginning of the simulation.
- Vehicles are moved to the nearest available intersection according to this map, with the rerouting time cost added to their waiting time.
- If no rerouting path exists, vehicles remain blocked until the accident or closure ends.

Enabling Rerouting

- The rerouting map must be defined at the beginning of the simulation. It is placed in the **Input File section** before the list of events. The map can be represented as an adjacency list or matrix, where each intersection lists its direct connections. This allows the simulator to compute alternate paths when required.
- For example, the input file may include a block after the intersection count specifying adjacency, such as:

```
1: 2 3
2: 1 4
3: 1 4
4: 2 3
```

This means intersection 1 connects to 2 and 3, and so on.

- Rerouting can be toggled in the input file with an extra configuration line (e.g., `Rerouting = ON` or `Rerouting = OFF`).
- If rerouting is ON, the simulator simply moves affected vehicles to the first available connected intersection as listed in the adjacency map. The additional travel can be modeled by adding a fixed waiting time penalty (e.g., +2 timesteps).
- If rerouting is OFF, vehicles remain stuck at their current lane or intersection until it reopens.

Event Queue Structure

- Events are sorted by time of occurrence.
- If multiple events occur at the same timestep, the order of execution is:
    1. Cancellations
    2. Promotions
    3. Accidents and Road Closures
    4. Arrivals
    5. Lane assignment and green signal update

This ensures that blocking or removing events are applied before new arrivals, and lane scheduling happens after the system state is fully updated.

Input / Output File Formats

1. Input File

- Line 1: Number of intersections I
- Line 2: Switching cost (timesteps)
- Line 3: AutoP threshold (timesteps)
- Line 4: CancelT threshold (timesteps)
- Line 5: Rerouting flag (ON/OFF)
- Line 6: Rerouting map (adjacency list of intersections)
- Line 7: Number of events E
- Next E lines = events (A, X, P, ACC, RC)

Sample Input File

```
4
2
5
10
ON
Connections:
1: 2 3
2: 1 4
3: 1 4
4: 2 3
8
A NC 1 101 1 1 3
A PT 2 201 2 1 4
A EV 3 301 1 2 2 5
ACC 4 1 1 3
P 5 201
X 6 101
A FV 7 401 3 1 5
RC 8 2 2
```

2. Output File

Each line:

CT ID AT WT XD TYPE INT LN [Status]

Each line corresponds to one vehicle and includes the following fields separated by spaces:

- **CT (Completion Time):** The timestep when the vehicle has completely crossed.
- **ID:** Unique identifier of the vehicle.
- **AT (Arrival Time):** Timestep the vehicle entered the system.
- **WT (Waiting Time):** How long the vehicle waited before starting to cross.
- **XD (Crossing Duration):** How long the crossing took.
- **TYPE:** The type of vehicle (EV, PT, NC, FV).
- **INT:** Intersection ID where the crossing occurred.
- **LN:** Lane ID within the intersection.
- **Status [Optional]:** If the vehicle was canceled, the record should clearly indicate **CANCELED** instead of CT by adding a (- -) sign for the CT part and adding **CANCELED** at the end.

At the end of the file, the system appends overall statistics that summarize performance:

- **Total vehicles:** By type (EV, PT, NC, FV) and overall.
- **Average waiting times:** Both overall and broken down by type.
- **Average crossing duration (XD).**
- **% of auto-promoted PT vehicles.**
- **Number of signal switches.**
- **Intersection utilization:** Percentage usage per intersection.
- **% of vehicles canceled.**

Sample Output File

```
CT ID AT WT XD TYPE INT LN
6 301 3 1 2 EV 1 2
7 201 2 3 4 EV 2 1
-- 101 1 5 3 NC 1 1 CANCELED
12 401 7 5 5 FV 3 1

Total Vehicles: 4
Total EV: 2 (including promotions)
Total PT: 0 (1 promoted)
Total NC: 1 (1 canceled)
Total FV: 1
Average WT (all): 3.0
Average WT (EV): 2.0
Average WT (NC): 5.0
Average XD: 3.5
% Auto-Promoted PT: 100%
Signal Switches: 3
% Vehicles Canceled: 25%
```